Agent Pipeline Development

Mehanički iznuđeni multi-agent pipeline sa hardverski postavljenim zaštitama.
Distribuira se kao Claude Code plugin — radi i na Codexu. Model-agnostic.

v4.7.21 · Zoran Stević · apd.run

PROBLEM

LLM piše kod.
Ko proverava da je pipeline ispoštovan?

Svaki tim "ima proces":
spec → plan → implementacija → review → testovi → commit.

U realnosti, kada LLM radi sam — proces je honor system.
Nema mehanizma da zaustavi agenta koji:

- preskoči spec i krene direktno u kod
- piše tamo gde mu nije dozvoljeno
- commituje bez testova i bez review-a
- izmeni `.done` fajl i pravi se da je sve uredu

CENTRALNA HIPOTEZA

Bez mehaničkog enforcementa, proces je sugestija.

Konvencija koju slabiji model u lošijem dan-u neće poštovati — i ti to nećeš primetiti dok ne pukne u produkciji.

• REŠENJE

APD = egzoskelet za LLM developera.

7-faza pipeline. 22 guard skripte koje blokiraju svaki bypass pokušaj. HMAC potpisi na `.done` fajlovima. Sve preživi compactization, restart sesije, i model switch.

MEHANIZAM

Hook-based enforcement

Svaki write, bash, commit prolazi kroz guard koji exit-uje 2 ako detektuje violation. Nema honor system-a.

DISTRIBUCIJA

Plugin, ne framework

`/plugin install` i sve je tu. Radi na Claude Code + Codex. Upstreamni plug-and-play.

DOKAZANO

33 logged runs

19/33 zero-block. 0 bypass pokušaja prošlo. Cela telemetrija audit-trail-uje se automatski.

DEO I

Pipeline mehanika.

7 faza. Svaki korak blokiran hook skriptom. Nema bypass-a iznutra Claude Code-a.

PIPELINE · 7 FAZA

Svaka faza. Svoj guard.

S	→	P	→	B	→	R	→	A	→	V	→	✓
Spec		**Plan**		**Builder**		**Reviewer**		**Adversarial**		**Verifier**		**Commit**
≤7 R-kriterija SHA256 freeze		plan.done signed before builder		scope locked guard-bash-scope		standard pass reviewer-order		context-free zero spec knowledge		@trace R* markers verify-all.sh		4×.done HMAC tamper-proof

Lean vs Full

- **Full (default):** svih 5 gate-ova + adversarial pass
- **Lean:** preskače adversarial kada je spec ≤2 R-kriterija
- Lean opt-out je mehanički: `adversarial: skip — <reason>` u spec-card

Reset je ekspliciatan

- Arhivira `agent-history.log` i `pipeline-metrics.log`
- Piše session-log entry sa "New rule" poljem
- **NIJE auto-triggered** — orchestrator mora pozvati (v4.7.21+)

PIPELINE STATE · .APD/PIPELINE/

Audit-trail koji ne možeš falsifikovati.

FAJL	UPISUJE	ŠTA ČUVA
spec-card.md	spec phase	frozen task spec
.spec-hash	spec phase	SHA256 tamper-guard
implementation-plan.md	orchestrator	fajlovi + agenti
*.done (×4)	svaki advance step	HMAC-signed markers
.reviewed-files	reviewer	reviewer scope
.adversarial-summary	adversarial	T:A:D + Notes ≥80 char
.trace-summary	verifier	covered/total + gaps
.agents	track-agent hook	pipe-delimited dispatch log

HMAC-signed .done format

```
# Line 1: metadata
1745510412|2026-04-18 14:20:12|fix-mailkit-bump

# Line 2: HMAC signature
a7f3c8b2e19d4f6a...
```

Potpisivanje radi validate-agent — Go binary čiji source **nije u repo-u.**

Ako orchestrator pročita pravila validacije, može ih zaobići. Zato source ostaje van repo-a, binary se distribuira za 4 platforme (darwin/linux × amd64/arm64).

SPEC TRACEABILITY · VERIFY-TRACE

Acceptance kriterij bez testa? Commit blokiran.

spec-card.md

```
## Acceptance Criteria

R1. Korisnik moze da postavi 6-cifreni OTP
R2. OTP istice nakon 10 minuta
R3. Pogresan OTP rate-limituje 5 pokusaja
R4. Email template renderuje u Gmail dark mode
R5. Mobile reset screen ima accessibility label
```

test-files / @trace markers

R1	6-digit OTP set	✓ otp_test.cs:24
R2	OTP expires 10min	✓ otp_test.cs:89
R3	Rate-limit 5 tries	✓ ratelimit_test.cs:12
R4	Gmail dark mode	✓ template_test.cs:203
R5	A11y label	⚠ NO @trace

```
$ bash bin/core/verify-trace TRACE:4/5:R5 ✗ Commit blocked — R5 has no @trace marker
```

DEO II

Mehanički enforcement.

22 guard skripte. Svaka odgovara na PreToolUse event. Svaki block loguje u guard-audit.log.

CORE GUARDS · BIN/CORE/GUARD-*

10 jezgrenih guard skripti.

guard-git

guard-git

Blokira `--no-verify`, force-push, `git reset --hard`, commits bez APD env-a

guard-bash-scope

guard-bash-scope

Bash write-ovi van agent scope-a: `cp`, `mv`, `>>`, runtime exec

guard-scope

guard-scope

Canonicalizes `realpath`; rejects `..`; prefix match allowed_paths

guard-orchestrator

guard-orchestrator

Orchestrator ne sme pisati code (stack-aware patterns). Samo .md, configs, docs/

guard-pipeline-state

guard-pipeline-state

Blokira write na `.done`, `.spec-hash`, `.trace-summary`, `verified.timestamp`

guard-lockfile

guard-lockfile

Basename whitelist: package-lock, composer.lock, Gemfile.lock, Cargo.lock, go.sum

guard-secrets

guard-secrets

Pattern list: `.env.production`, `.pem`, `id_rsa`, user-secrets

guard-send-message

guard-send-message

Blokira SendMessage kada spec.done postoji — bypasses SubagentStart hook

guard-compact

guard-compact

PreCompact — blokira compact ako pipeline ima active `.done` fajlove

guard-permission-denied

guard-permission-denied

Telemetrija. Loguje permission-denied u audit. Nikad ne blokira.

Svaki block log-uje `log_block` helper → `<memory>/guard-audit.log`

Pipe-delimited: ts | TYPE | agent_info | reason | cmd_summary

DIAGNOSTIC · APD DOCTOR

11-sekcionski audit pipeline-a.

apd doctor — zsh

```
$ apd doctor ── APD Doctor · v4.7.21 ── ✓ 1. Pipeline state spec.done builder.done reviewer.done verifier.done ✓ 2. Spec card spec-card.md present ✓ 3. Spec freeze SHA256 matches .spec-hash ✓ 4. Implementation plan implementation-plan.md present ✓ 5. Agent registry 5 agents · all frontmatter valid ✓ 6. Guard executability 22/22 chmod +x ✓ 7. Trace coverage 7/7 R-criteria covered ✓ 8. Adversarial summary 3/3 real, 0 dismissed ⚠ 9. GitHub sync issue #127 not closed ✓ 10. Plugin version v4.7.21 · compatible ✓ 11. Legacy paths none detected ── OK 10 passed · 1 warning · 0 failed · exit 0
```

REZULTAT · APD REPORT

Ljudima-čitljiv recap svakog runa.

apd report — #33 · MailKit STARTTLS bump

```
┌──────────────
```

DEO III

Runtime contract.

Jedan pipeline. Dva LLM runtime-a. Adapter arhitektura — ne copy-paste.

ADR-001 · RUNTIME CONTRACT

3-sloj arhitektura.

Komponente CORE

pipeline-advance	CORE
validate-agent	CORE
verify-trace	CORE
verify-all.sh	CORE
guard-scope	MIXED
guard-orchestrator	MIXED
hooks.json (CC)	ADAPTER
MCP server (Codex)	ADAPTER

Problem koji rešava

9 guarda je direktno parsiralo CC hook stdin JSON.
Ako se CC protocol promeni — sve puca istovremeno.

Odluka

- Definiši **APD Runtime Contract** — 7 normalizovanih event-a
- Klasifikuj: Core · Mixed · Adapter
- Migracija u 3 faze (Phase 1+2 gotove)

> CC update ne ruši core logiku. Core je testabilan bez CC-a. Drugi runtime (Codex) pristupa istom core-u kroz svoj adapter.

DISTRIBUCIJA

Claude Code · Codex · Hybrid.

LAYER	CLAUDE CODE	OPENAI CODEX
Plugin manifest	.claude-plugin/plugin.json v4.7.21	plugins/apd/.codex-plugin/ v4.7.20 (lags)
Marketplace	zstevovich-plugins	codex-apd · INSTALLED_BY_DEFAULT
Install	/plugin install	upstream-blocked · direct-drop
Hook engine	13 hook entries · 7 event types	1 hook · PreToolUse Bash (only reliable)
Agent dispatch	Native SubagentStart/Stop	Inline + MCP tool
Enforcement gate	hooks + validate-agent	MCP server (8 tools)
Skills	8 (full)	4 (subset) + inline via apd cdx

CODEX · MCP/APD_MCP_SERVER.PY

8 MCP alata = Codex enforcement.

TOOL	PURPOSE
apd_ping	Liveness + version
apd_doctor	In-conversation diagnostics
apd_pipeline_state	Strukturovan snapshot
apd_list_agents	Registry sa frontmatter
apd_advance_pipeline	Whitelist: spec/builder/reviewer/verifier/…
apd_guard_write	Pre-write auth + regex + FS escape
apd_verify_step	verify-all.sh sa scope env
apd_adversarial_pass	Notes ≥80 char when total=0

Defense in depth — apd_guard_write

```
# 1. Regex whitelist za apd_role
ROLE_RE = r'^[A-Za-z0-9_.-]+$'

# 2. Filesystem escape defense
agent_file = agents_dir / f"{role}.md"
assert agent_file.parent.resolve() == agents_dir

# 3. readonly coercion
if bool(parsed.get("readonly", False)):
    return {"exit_code": 2, "error": "readonly agent"}

# 4. scope kao lista — empty ako nedostaje
scope = parsed.get("scope", []) or []

# 5. Delegiraj na core/guard-scope
return _run_core("guard-scope", ...)
```

AGENT REGISTRY

Agenti su scope-ovani u frontmatter-u.

Frontmatter (Builder agent)

```yaml
---
name: backend-api
description: Backend builder
tools: Read, Write, Edit, Bash
model: sonnet
effort: xhigh      # ≈ 40 turns
maxTurns: 40
memory: project
permissionMode: bypassPermissions
readonly: false
scope:
  - src/Api/
  - src/Services/
  - tests/Api/
---
```

5 default agent template-a

AGENT	MODEL	TURNS
backend-builder	sonnet	40
frontend-builder	sonnet	40
testing	sonnet	30
code-reviewer	opus	30
adversarial-reviewer	opus	30

KRITIČNO

Scope server-side čita iz agent frontmatter-a. Caller-supplied allowed_paths se ignoriše — agent ne može lagati o svom scope-u.

DEO IV

Rezultati.

33 logged runs. 0 bypass pokušaja prošlo. Model-agnostic dokaz koji je boleo.

PROIZVODNA TELEMETRIJA

Cifre iz `pipeline-metrics.log`.

33

logged pipeline runs

6 projekata · .NET + PHP + GLM-5

19/33

zero-block runs

58% clean · nijedan guard triggered

22

guard skripte

sve aktivne · auditable · exit 2 = block

0

bypass pokušaja prošlo

52 pokušaja na GLM-5 · svi blokirani

8

MCP tool endpoint-a

Codex runtime · per-tool approval

100%

spec coverage

7/7 R-kriterija → test files (@trace)

MODEL-AGNOSTIC CASE · RUN #21

Isti pipeline. Drugi model.
Isti rezultat.

TinyMCE → TipTap migracija — 7 React komponenata, 99 fajlova, 1349 insertions, 2208 deletions.

METRIKA	Claude Opus	GLM-5	DELTA
Pipeline time	~10 min	17m 13s	1.7×
Completion rate	100%	100%	match
Guard blocks	1–2	52	25×
Spec coverage	7/7	7/7	match
Builder iterations	0–1	3+	3×
Adversarial findings	3–5	10 (3 real)	–

KLJUČNI NALAZ

GLM-5 je pokušao direktan code-write (blokirao guard-orchestrator), commit bez pipeline-a (blokirao pipeline-

DEO V

60 sekundi do prvog run-a.

Tri komande. Jedan plugin. Nema reboot-a, nema konfiguracije.

INSTALACIJA · CLAUDE CODE

Tri komande.

STEP 01

Dodaj marketplace

```
/plugin marketplace add zstevovich/claude-apd Fetching plugin metadata… ✓ Added zstevovich-plugins
```

STEP 02

Instaliraj plugin

```
/plugin install claude-apd@zstevovich-plugins Installing v4.7.21… → 22 hooks · 8 skills · 5 agents ✓ Plugin active
```

STEP 03

Scaffold + verifikuj

```
/apd-setup Scaffolding .claude/ tree… ✓ .claude/bin/, hooks/, skills/, agents/ ✓ stack: .NET · project: MyApp $ bash .claude/bin/apd verify —— APD self-check —— ✓ pipeline-gate.sh OK ✓ guard-bash-scope.sh OK ✓ verify-trace.sh OK ✓ adversarial-gate.sh OK … 18 more … ✓ all 22 guards ready
```

Codex setup: `apd cdx init` + `apd cdx skills install`

STACK TEMPLATES · VERIFY-ALL.SH

6 stack-ova · per-project scaffold.

STACK	BUILD	TEST	CONFIG
node	npm build	npm test	package.json · separate frontend if present
python	pip install	pytest	requirements.txt · coverage
php	composer install	phpunit	composer.json
dotnet	dotnet build	dotnet test	.csproj · NuGet
go	go build	go test	go.mod · go.sum
java	mvn / gradle	+ test suite	pom.xml / build.gradle

Svaki template šalje CHANGED_FILES preko git diff i kešira verifier rezultat 120s.
APD_VERIFY_SCOPE=fast — samo touched testovi. APD_VERIFY_SCOPE=full — cela suite.

KNOWN LIMITATIONS & BACKLOG

Šta još ne radi.

Codex upstream

- Codex 0.121.0 marketplace install blokiran — openai/codex#18258
- `/` slash menu ne lista APD skills
- Codex sandbox profiles disabled (runtime ne poštuje FileSystemSandboxPolicy)

Runtime gaps

- `SessionStart` hook ne okida na nekim projektima (CRITICAL backlog)
- `guard-parallel-same-agent` nedostaje — 3× parallel dispatch = conflicts
- CC `SubagentStop` je jedini telemetry source na CC-u

Sigurnosni backlog

- `mkdir .pipeline/` orchestrator bypass — fix via permissions.deny
- Superpowers plugin conflict — mehanički blokovan
- In-monorepo `verify-apd` mis-resolves project root

Drift

- Codex plugin manifest lags CC — v4.7.20 vs v4.7.21
- Pipeline reset **NIJE** auto-triggered — orchestrator mora pozvati
- ADR-001 Phase 3 (second runtime adapter) — in-progress

ROADMAP · NAREDNA 4 RELEASE-A

Gde ide APD.

V4.8 · NAREDNI RELEASE

Runtime Contract Phase 3

Druga runtime adapter implementacija — Codex 0.130.0+ kada upstream bug bude zatvoren. Skidanje svakog CC-specifičnog parse koda iz guards.

V4.9

Parallel-dispatch guard

guard-parallel-same-agent — blokira više istih agenata u paralelnom dispatch-u. Trenutno conflict može da prebriše state.

V5.0

Evidence-based adversarial

Workflow rule iz Run #32 formalizovana: adversarial findings sa HIGH severity moraju imati WebFetch dokaz pre dismissa-a.

V5.1

Cross-project learning

MEMORY.md sinhronizacija preko projekata — skip-log pattern detection. Gde često pravimo spec-preskakanje = problem u procesu.

PITANJA · DISKUSIJA · HANDS-ON

APD je open source.
Probajte sami.

```
/plugin marketplace add zstevovich/claude-apd /plugin install
claude-apd@zstevovich-plugins /apd-setup
```

apd.run · github.com/zstevovich/claude-apd · @zstevovich